

CO2 - ASSESSMENT TOOL- 2 -SCENARIO BASED QUESTIONS

CODE: DSA0108-OBJECT ORIENTED PROGRAMMING WITH C++

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Q1. Hospital Consultation Charge System

```
#include <iostream>

#include <string>

using namespace std;

double calculateCharge(double generalFee);

double calculateCharge(double specialistFee, int specialist);

void modifyBill(double &bill);

double addServiceCharge(double bill, double serviceCharge = 100);

int main()
{
    int choice;

    double bill;

    string patientType;

    cout << "1. General Consultation\n";

    cout << "2. Specialist Consultation\n";

    cout << "Enter Choice: ";

    cin >> choice;

    if(choice == 1)
    {
        patientType = "General";

        double fee;

        cout << "Enter General Consultation Fee: ";

        cin >> fee;

        bill = calculateCharge(fee);

    }
```

```

else
{
    patientType = "Specialist";
    double fee;
    cout << "Enter Specialist Consultation Fee: ";
    cin >> fee;
    bill = calculateCharge(fee,1);
}
bill = addServiceCharge(bill);
modifyBill(bill);
cout << "\nPatient Type : " << patientType;
cout << "\nTotal Consultation Charge : " << bill;
return 0;
}

double calculateCharge(double generalFee)
{
    return generalFee;
}

double calculateCharge(double specialistFee, int specialist)
{
    return specialistFee + 300;
}

void modifyBill(double &bill)
{
    bill = bill - 50;
}

double addServiceCharge(double bill, double serviceCharge)
{
    return bill + serviceCharge;
}

```

```
}
```

OUTPUT:

```
Output
1. General Consultation
2. Specialist Consultation
Enter Choice: 1
Enter General Consultation Fee: 1500

Patient Type : General
Total Consultation Charge : 1550

=== Code Execution Successful ===
```

Q2. Fitness Centre Membership System

```
#include <iostream>

#include <string>

using namespace std;

class Member
{
    int id;

    string name;

    string plan;

    double fee;

public:

    static int activeMembers;

    void input()
    {
        cout<<"Enter ID: ";

        cin>>id;
```

```

cout<<"Enter Name: ";
cin>>name;
cout<<"Enter Plan (Basic/Premium/VIP): ";
cin>>plan;

if(plan=="Basic")
    fee=1000;
else if(plan=="Premium")
    fee=2000;
else if(plan=="VIP")
    fee=3000;
else
{
    cout<<"Invalid Plan\n";
    fee=0;
}
activeMembers++;
}

void display()
{
    cout<<"\nID : "<<id;
    cout<<"\nName : "<<name;
    cout<<"\nPlan : "<<plan;
    cout<<"\nFee : "<<fee<<endl;
}

static void totalMembers()
{
    cout<<"\nActive Members : "<<activeMembers<<endl;
}

```

```

}

int Member::activeMembers=0;

int main()
{
    int n;

    cout<<"Enter Number of Members: ";

    cin>>n;

    Member m[20];

    for(int i=0;i<n;i++)
    {

        cout<<"\nMember "<<i+1<<endl;

        m[i].input();

    }

    cout<<"\nMembership Details\n";

    for(int i=0;i<n;i++)
    {

        m[i].display();

    }

    Member::totalMembers();

    return 0;

}

```

Output:

```

Enter Number of Members: 2
Member 1
Enter ID: 123
Enter Name: TEJU
Enter Plan (Basic/Premium/VIP): VIP
Member 2
Enter ID: 1234
Enter Name: HARSHII
Enter Plan (Basic/Premium/VIP): Baic
Invalid Plan
Membership Details
ID : 123
Name : TEJU
Plan : VIP
Fee : 3000
ID : 1234
Name : HARSHII
Plan : Baic
Fee : 0
Active Members : 2

```

Q3. Loan Repayment System

```
#include <iostream>

using namespace std;

double calculateEMI(double loan, int years, double rate = 8.5);

inline double calculateInterest(double loan, double rate);

void applySubsidy(double &amount);

int main()
{
    double loan, emi, interest;

    int years;

    cout << "Enter Loan Amount: ";

    cin >> loan;

    cout << "Enter Loan Period (Years): ";

    cin >> years;

    interest = calculateInterest(loan, 8.5);

    emi = calculateEMI(loan, years);

    applySubsidy(loan);

    cout << "\nLoan Amount after Subsidy : " << loan;

    cout << "\nInterest : " << interest;

    cout << "\nMonthly EMI : " << emi;

    return 0;
}

inline double calculateInterest(double loan, double rate)
{
    return (loan * rate) / 100;
}

double calculateEMI(double loan, int years, double rate)
{
    double total = loan + (loan * rate * years / 100);
```

```

        return total / (years * 12);
    }

    // Call by Reference
    void applySubsidy(double &amount)
    {
        amount = amount - 10000;
    }

```

output:

```

Enter Loan Amount: 1000
Enter Loan Period (Years): 5

Loan Amount after Subsidy : -9000
Interest : 85
Monthly EMI : 23.75

=== Code Execution Successful ===

```

Q4. Parcel Charge Management System #include <iostream>

```

#include <string>

using namespace std;

class Parcel
{
private:
    string type;
    float weight;
    float distance;
    float packageCharge;
public:
    static int totalParcels;
    void readData();
    inline float baseCharge();
    float calculateCharge(float packaging = 50);
    float calculateCharge(int express, float packaging = 50);

```

```

void display();

static void showTotal()
{
    cout << "\nTotal Parcels Processed : " << totalParcels << endl;
}

};

int Parcel::totalParcels = 0;

void Parcel::readData()
{
    cout << "Enter Parcel Type (Normal/Express): ";
    cin >> type;
    cout << "Enter Weight (kg): ";
    cin >> weight;
    cout << "Enter Distance (km): ";
    cin >> distance;
    totalParcels++;
}

inline float Parcel::baseCharge()
{
    return weight * 20 + distance * 5;
}

float Parcel::calculateCharge(float packaging)
{
    packageCharge = baseCharge() + packaging;
    return packageCharge;
}

float Parcel::calculateCharge(int express, float packaging)
{
    packageCharge = baseCharge() + packaging + 100;
}

```



```

        return packageCharge;
    }

void Parcel::display()
{
    cout << "\nParcel Type : " << type;
    cout << "\nWeight : " << weight;
    cout << "\nDistance : " << distance;
    cout << "\nTotal Charge : " << packageCharge << endl;
}

int main()
{
    Parcel p;
    p.readData();
    char ch;
    cout << "Express Delivery? (y/n): ";
    cin >> ch;
    if(ch=='y' || ch=='Y')
        p.calculateCharge(1);
    else
        p.calculateCharge();
    p.display();
    Parcel::showTotal();
    return 0;
}

```

Output

```
Output
Enter Parcel Type (Normal/Express): Normal
Enter Weight (kg): 14
Enter Distance (km): 15
Express Delivery? (y/n): y

Parcel Type : Normal
Weight : 14
Distance : 15
Total Charge : 505

Total Parcels Processed : 1

=== Code Execution Successful ===
```

Q5. Water Billing System

```
#include <iostream>

#include <string>

using namespace std;

// Function Prototypes
void applySubsidy(double &bill);

class Consumer
{
private:
    int id;
    string type;
    double units;
    double bill;
public:
    static int totalConsumers;
    void input();
    void calculate(double tax = 50);
    void calculate(int commercial, double tax = 50);
    void display();
    static void showCount()
    {
        cout << "\nTotal Consumers : " << totalConsumers << endl;
```

```

    }
};

int Consumer::totalConsumers = 0;

void Consumer::input()
{
    cout << "Enter Consumer ID: ";
    cin >> id;
    cout << "Enter Consumer Type (Domestic/Commercial): ";
    cin >> type;
    cout << "Enter Water Consumption (Litres): ";
    cin >> units;
    totalConsumers++;
}

void Consumer::calculate(double tax)
{
    bill = units * 2 + tax;
}

void Consumer::calculate(int commercial, double tax)
{
    bill = units * 5 + tax;
}

void Consumer::display()
{
    cout << "\nConsumer ID : " << id;
    cout << "\nConsumer Type : " << type;
    cout << "\nWater Consumption : " << units;
    cout << "\nBill Amount : " << bill << endl;
}

```

```

void applySubsidy(double &bill)
{
    bill = bill - 100;
}

int main()
{
    int n;
    cout << "Enter Number of Consumers: ";
    cin >> n;
    Consumer c[20];
    for(int i=0;i<n;i++)
    {
        cout << "\nConsumer " << i+1 << endl;
        c[i].input();
        char ch;
        cout << "Commercial Consumer? (y/n): ";
        cin >> ch;
        if(ch=='y' || ch=='Y')
            c[i].calculate(1);
        else
            c[i].calculate();
        c[i].display();
    }
    Consumer::showCount();

    return 0;
}

```

Output:

```
Enter Number of Consumers: 2

Consumer 1
Enter Consumer ID: 1234
Enter Consumer Type (Domestic/Commercial): Domestic
Enter Water Consumption (Litres): 1
Commercial Consumer? (y/n): y

Consumer ID : 1234
Consumer Type : Domestic
Water Consumption : 1
Bill Amount : 55

Consumer 2
Enter Consumer ID: 1222
Enter Consumer Type (Domestic/Commercial): Commercial
Enter Water Consumption (Litres): 2
Commercial Consumer? (y/n): n

Consumer ID : 1222
Consumer Type : Commercial
Water Consumption : 2
Bill Amount : 54

Total Consumers : 2

=== Code Execution Successful ===
```